

RBG-48377

10 CFR 50.73

October 20, 2025

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2025-006-00, Potential Loss of Safety
Function of Control Building Chillers Due to a Valve Mispositioned

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report.

This document contains no commitments.

Should you have any questions, please contact Mr. Jack McCoy, Regulatory and Emergency Planning Manager, at 225-381-3310.

Respectfully,

**Jack
McCoy** 
Digitally signed by Jack McCoy
DN: cn=Jack McCoy, o=Manager
Regulatory and Emergency Preparedness,
ou=Entergy/ River Bend ,
email=jmccoy1@entergy.com
Date: 2025.10.20 10:46:56 -05'00'

JM/hpk

Enclosure: Licensee Event Report 50-458 / 2025-006-00

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station
NRC Office of Nuclear Security and Incident Response

Enclosure

RBG-48377

Licensee Event Report
50-458/2025-006-00

NRC FORM 366 (04-02-204)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.					
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)								
1. Facility Name River Bend Station, Unit 1					☒ 050 ☐ 052		2. Docket Number 00458		3. Page 1 of 3	
4. Title Potential Loss of Safety Function of Control Building Chillers Due to a Valve Mispositioned										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050
8	23	2025	2025	006	00	10	20	2025	N/A	Docket Number
						Facility Name N/A				
						Facility Name N/A				
9. Operating Mode						10. Power Level				
1						100				
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)(i)		☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)				
☐ Other (Specify here, in abstract, or in NRC 366A).										
12. Licensee Contact for this LER										
Licensee Contact Jack McCoy								Phone Number (Include Area Code) (225) 381-3310		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS	
A	VI	CHU	W120	Y						
14. Supplemental Report Expected								15. Expected Submission Date		
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)								Month Day Year		
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)										
On August 23, 2025, at 1433 CDT, River Bend Station, Unit 1 (RBS) was operating in Mode 1 at 100% reactor power, a post maintenance test was being performed on Division II Control Building Chilled Water System Chiller B (HVK-CHL1B). During this test, Division I Control Building Chilled Water System (HVK) was declared Inoperable as a result of HVK Chiller A (HVK-CHL1A) failing to start. HVK-CHL1A failed to start due to the HVK Low Side Isolation Valve, HVK-FTY5B-V2L, being in the closed position. The closed isolation valve prevented a "low flow" signal from the transmitter and render Division I of the HVK Chillers, HVK-CHL1A and HVK-CHL1C, inoperable. Operations immediately alternated back to HVK-CH1B, with HVK-CH1C in standby. This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(A) and 10 CFR 50.73(a)(2)(v)(D) as an event that could have prevented the fulfillment of the safety function of structures or systems that are needed to shut down the reactor and maintain it in a safe shutdown condition and mitigate the consequences of an accident.										

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station, Unit 1	☒ 050	00458	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	006	00

NARRATIVE**Event Description**

On August 23, 2025, at 1433 CDT, River Bend Station, Unit 1 (RBS) was operating in Mode 1 at 100% reactor power, a post maintenance test was being performed on Division II Control Building Chilled Water System Chiller B (HVK-CHL1B) [VI:CHU]. During this test, while transferring operation from HVK Chiller B (HVK-CHL1B) to HVK Chiller A (HVK-CHL1A), HVK-CHL1A failed to start. Division I Control Building Chilled Water System (HVK) was declared Inoperable. The inoperability of Division I and II Control Building Chilled Water Systems could have prevented the fulfillment of a safety function.

HVK-CHL1A failed to start due to the HVK Low Side Isolation Valve, HVK-FTY5B-V2L, being in the closed position. The closed isolation valve prevented a "low flow" signal from the transmitter and render Division I of the HVK Chillers, HVK-CHL1A and HVK-CHL1C, inoperable. Operations immediately alternated back to HVK-CH1B, with HVK-CH1C in standby.

Limiting Conditions for Operation (LCO) 3.7.3, Control Room Air Conditioning System Condition A (One control room AC subsystem inoperable) and 3.7.7, Control Building Air Conditioning System Condition A (one control building air conditioning subsystem inoperable) was entered.

After further investigation, it was determined that Instrument and Control (I&C) Technicians manipulated HVK-FTY5B-V2L to the closed position earlier that day during performance of a work order to fill and vent transmitters on the Control Building Chiller 1B. The I&C Technicians failed to document the change of valve position as required by EN-MA-157, Maintenance Configuration Control, therefore the valve was not returned to its proper position (i.e., open).

Event Cause

The direct cause of this event was due to the misposition of HVK-FTY5B-V2L that resulted in the valve being left in the closed position by I&C Technicians.

The causal factor was determined to be that the workers failed to stop work and notify supervision when a potential issue was encountered. The workers did not utilize EN-MA-157, Maintenance Configuration Control, before manipulating a valve. The workers failed to identify potential error traps and failed to utilize human performance tools as required by EN-HU-102, Human Performance Traps & Tools.

Safety Assessment

Dedicated operator actions had been established to maintain the safety function for the HVK system in the event that the standby chiller doesn't start and the previously running chiller doesn't start when alternating divisions of control building chilled water as a planned evolution. Safety function is maintained if a chiller is started within 23 minutes, which is the Environmental Design Criteria limitation. The safety significance of this event is low based on temperatures in the areas cooled by the HVK systems remained below maximum environmental design criteria and actions taken by Operations.

Corrective Actions**Completed:**

- Maintenance Department held a stand-down for a standards reset.
- Maintenance Department maintained 100% Supervisor job oversight for 2 weeks following the event.



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In Progress:

- Performance management actions will be used to reinforce the behaviors with the individuals to prevent recurrence.
- Performance of observations to ensure human performance standards for procedure use and adherence are being observed and coached.

Previous Occurrences

One similar event was reported in LER 050/458-2020-001-00. The plant operating at 100 percent power, a post modification test was performed on Division I Control Building Chilled Water System Chiller A (HVK-CHL1A). During the test, both Division I and Division II Control Building Chilled Water Systems (HVK) were declared Inoperable resulting in a condition that could have prevented fulfillment of a safety function. The event occurred while transferring from operation of HVK-CHL1B to HVK-CHL1A as part of the test. The failure of HVK-CHL1A was attributed to a trip on a Phase Reversal alarm within the digital control logic, during post modification testing.

NOTE: Energy Industry Identification System component function identifier and system name of each component or system referred to in the LER are annotated as (**XX**) and [XX], respectively.